The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 September 1997

INCH-POUND

MIL-PRF-19500/486E
10 June 1997
SUPERSEDING
MIL-S-19500/486D
27 June 1995

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, COUPLER, OPTOELECTRIC, SOLID STATE;
TYPES 4N22, 4N22A, 4N23, 4N23A, 4N24, 4N24A, 4N22U, 4N22AU,
4N23U, 4N23AU, 4N24U, 4N24AU; JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for solid state optically coupled isolators in which a gallium arsenide or gallium aluminum arsenide diode light source is optically coupled to a silicon NPN phototransistor. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1 and 2.

1.3 Maximum ratings. Unless otherwise specified, maximum ratings apply to all case outlines and T_A = +25°C.

1.3.1 Infrared-emitting diode maximum ratings.

V_R	I_F 1/	I_P 2/
V dc	mA dc	A (pk)
2	40	1

1/ Derate linearly to +125°C at 0.67 mA/°C above +65°C.
2/ 1.0 µs pulse width, 300 pps.

1.3.2 Phototransistor maximum ratings.

V_{CEO}	V_{CBO}	V_{EBO}	P_T 1/	I_C
V dc	V dc	V dc	mW	mA dc
35	35	4	300	50

1/ Derate linearly to +125°C at 3 mA/°C above +25°C.

1.3.1 Total device ratings.

T_J and T_{STG}	V_{IO}
°C	V dc
-65 to +125	1,000 max

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics. Unless otherwise specified, electrical characteristics apply to all case outlines and T_A = +25°C.

1.4.1 LED (input) characteristics.

Limits	I_R V_R = 2 V dc	V_F I_F = 10 mA dc
	A dc	V dc
Min		0.8
Max	100	1.3

1.4.2 Phototransistor (output) characteristics.

Limits	$I_{C(OFF)}$ V_{CE} = 20 V dc	$V_{(BR)CEO}$ I_C = 1 mA dc	$V_{(BR)CBO}$ I_C = 100 µA dc	$V_{(BR)CBO}$ I_E = 100 µA dc	h_{FE} V_{CE} = 20 V dc I_C = 10 mA dc		
					4N22 4N22A	4N23 4N23A	4N24 4N24A
	nA	V dc	V dc	V dc			
Min		35	35	4	200	300	400
Max	100						

1.4.2 Coupled (transfer) characteristics.

Limits	R_{IO} at 1,000 V dc see 4.5.4	C_{IO} V = 0 f = 1 MHz see 4.5.4	Phototransistor mode switching (see figure 3) I_C = 100 µA dc					
			4N22	4N22A	4N23	4N23A	4N24	4N24A
			t_r	t_f	t_r	t_f	t_r	t_f
	ohms	pF	µs	µs	µs	µs	µs	µs
Min	10^{11}							
Max		5	15	15	15	15	20	20

Limits	$V_{CE(SAT)}$ I_F = 20 mA			$I_{C(ON)2}$ I_F = 10 mA V_{CE} = 5 V dc			$I_{C(ON)1}$ I_F = 2 mA V_{CE} = 5 V dc		
	I_C = 2.5 mA	I_C = 5 mA	C = 10 mA						
	4N22 4N22A	4N23 4N23A	4N24 4N24A	4N22 4N22A	4N23 4N23A	4N24 4N24A	4N22 4N22A	4N23 4N23A	4N24 4N24A
	V dc	V dc	V dc	mA dc	mA dc	mA dc	mA dc	mA dc	mA dc
Min				2.5	6.0	10.0	0 15	0.2	0.4
Max	0.3	0.3	0.3						

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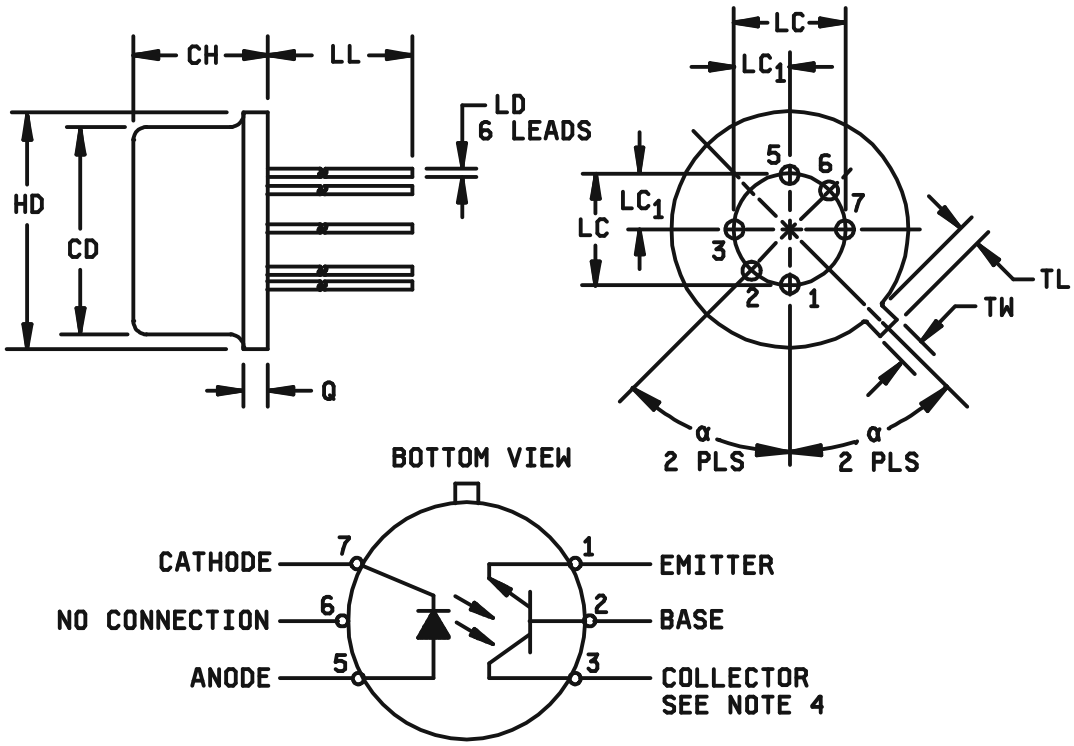


FIGURE 1. Physical dimensions and configuration.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
HD	.335	.370	8.51	9.40	5
CD	.305	.335	7.75	8.51	5
LD	.016	0.19	0.41	0.48	5
α	45° T.P.		45° T.P.		3
Q	- - -	.040	- - -	1.02	
CH	.155	.185	3.94	4.70	
LC	.200 T.P.		5.08 T.P.		3
LC_1	.100 T.P.		2.54 T. P.		3
TW	.028	.034	0.71	0.86	
TL	.029	.045	0.74	1.14	
LL	.500	.600	12.70	15.24	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. T.P. designates true position. Leads having maximum diameter .019 inches (0.48 mm) measured in gauging plane .054 + .001, -.000 inch (1.37 +0.03, -0.00 mm) below the seating plane of the device shall be within .007 inch (0.18 mm) of their true position relative to a maximum width tab.
4. 4N22A, 4N23A, and 4N24A have collector isolated from the case.
5. In accordance with ANSI Y14.5M, diameters are equivalent to ∅x symbology.

FIGURE 1. Physical dimensions and configurations - Continued.

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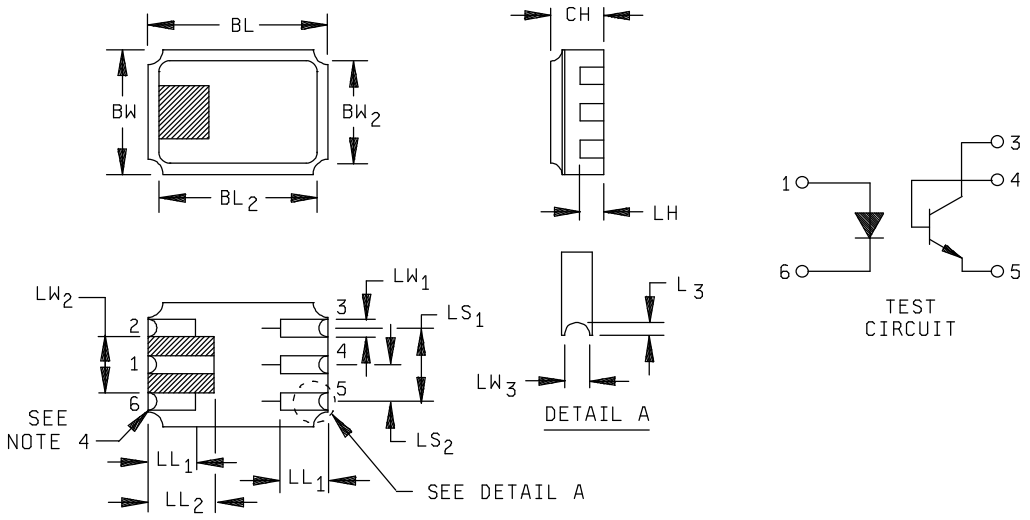


FIGURE 2. Physical dimensions and configuration, surface mount devices.

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Ltr	Dimensions				Notes
	Inches		Millimeter		
	Min	Max	Min	Max	
CH	.058	.080	1.47	2.03	
LH	.026	.034	0.66	0.86	
LW_1	.022	.028	0.56	0.71	
LW_2	.072 REF		1.83 REF		
LW_3	.006	.022	0.15	0.56	
BW	.165	.175	4.19	4.44	
LS_1	.095	.105	2.41	2.67	
LS_2	.045	.055	1.14	1.39	
BW_2	- - -	.175	- - -	4.44	
BL	.240	.250	6.10	6.35	
BL_2	- - -	.250	- - -	6.35	
LL_1	.060	.070	1.52	1.78	
LL_2	.082	.098	2.08	2.49	
L_3	.003	.007	0.08	0.18	
N	6				3
ND	3				3
NE	0				3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Symbol "N" is the maximum number of terminals. Symbols "ND" and "NE" are number of terminals along the length "BW" and "BL" respectively.
4. Pin number designation:
   1 - Anode. 3 - Collector. 5 - Emitter.
   2 - N/C. 4 - Base. 6 - Cathode.

FIGURE 2. Physical dimensions and configurations, surface mounted devices - Continued.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

$I_{C(ON)}$ ---------- On-state collector current.

$I_{C(OFF)}$ ---------- Off-state collector current.

T_T ---------- Temperature, terminal pad.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions for the purpose of interchangeability shall be as specified on figures 1 and 2 herein. With the approval of the qualifying activity, this device is authorized to contain organic material for JANS quality devices.

3.3.1 Lead finish. Lead finish shall solderable in accordance with MIL-PRF-19500, MIL-STD-750 and herein. Lead finish shall be gold plated, tin plated, or solder dipped. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

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3.3.2 For eutectic alloyed die of less than 1.5 x 10^{-5} square inches (0.10 mm^2) of surface area. Die shear process control shall be in accordance with the requirements for small area die in MIL-STD-883, method 2019.

3.3.3 JANTX and JANTXV only. Die shear process control on epoxy mounted die is not required.

3.4 Marking. Markign shall be as specified in MIL-PRF-19500.

3.4.1 Marking for surface mount devices. Marking shall be in accordance with MIL-PRF-19500.

3.5 Moisture content. The internal moisture content of the device package shall not exceed 10,000 PPM at +100°C.

3.6 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 2 and 3.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

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4.3 Screening (JANTX, JANTXV, and JANS levels). Screening shall be in accordance with appendix E, table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3	See 4.3.1	See 4.3.1
9 1/	$I_{C(OFF)}$, h_{FE}	$I_{C(OFF)}$, 100 percent read and record.
10	See 4.3.2	See 4.3.2
11 2/	I_R, $I_{C(OFF)}1$, h_{FE} and $I_{C(ON)}2$; $\Delta I_{C(OFF)}1$ = 100 percent of initial value or 25 nA dc, whichever is greater. Δh_{FE} = ± 20 percent of initial reading.	I_R, $I_{C(OFF)}1$, h_{FE} and $I_{C(ON)}2$; $\Delta I_{C(OFF)}1$ = 100 percent of initial value or 25 nA dc, whichever is greater.
12	See 4.3.3	See 4.3.3
13	Subgroup 2 of table I herein; $\Delta I_{C(OFF)}1$ = 100 percent of initial value or 25 nA dc, whichever is greater. Δh_{FE} = ± 20 percent of initial reading. $I_{C(ON)}$ = ± 25 percent of initial reading; ΔI_R = 100 percent of initial value or 25 µA dc, whichever is greater.	Subgroup 2 of table I herein; $\Delta I_{C(OFF)}1$ = 100 percent of initial value or 25 nA dc, whichever is greater. Δh_{FE} = ± 20 percent of initial reading. $I_{C(ON)}$ = ± 25 percent of initial reading; ΔI_R = 100 percent of initial value or 25 µA dc, whichever is greater

1/ Screen 9 may be omitted if $I_{C(OFF)}1$ in screen 11 is determined using the maximum limit of $I_{C(OFF)}1 \leq 25$ nA (see screen 11).
2/ If screen 9 was omitted, $I_{C(OFF)}1$ maximum limit is 25 nA and $\Delta I_{C(OFF)}1$ is not required.

4.3.1 Thermal shock (temperature cycling). All devices shall be subjected to thermal shock (temperature cycling) in accordance with MIL-STD-750, method 1051, test condition B, except $T_{(min)}$ = -55°C; 10 cycles, time at temperature extremes = 15 minutes minimum.

4.3.1.1 Monitored thermal shock (temperature cycling) (JANS only). One cycle on monitored cycling shall be performed on 100 percent of the devices. This test shall be performed any time after the completion of the thermal shock test specified or it may be the last of the 10 thermal cycles. All junctions shall be monitored for electrical continuity. Any discontinuity shall be cause for rejection of the device(s) under test. If 10 percent or more of the number of devices subjected to monitored temperature cycling fail, the entire lot shall be rejected as JANS types.

4.3.2 High temperature reverse bias (HTRB). All devices shall be subjected to high temperature reverse bias in accordance with MIL-STD-750, method 1039, test condition A, T_A = +125°C, I_F = 0, V_{CB} = 28 V dc for 48 hours minimum.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Burn-in circuit (see figure 4); V_{CC} = 20 V dc, V_{CE} = 10 V dc ± 5 V dc, I_F = 40 mA dc, P_T = 275 = ± 25 mW. For surface mount devices, T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750; T_T = +100°C to +125°C for surface mounted devices. No heat sink or forced air cooling on the devices shall be permitted.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500, and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIb (JAN and JANTX) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1051	Test condition B.
B4	1037	I_F = 20 mA dc; V_{CE} = 10 V dc; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. P_T = 275 ± 25 mW at T_A = room ambient as defined in the general requirements of MIL-STD-750. No heat sink or forced air cooling shall be permitted on the devices.

4.4.2.1 Group B inspection, appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	1051	Test condition B.
B3	1027	I_F = 20 mA dc; V_{CE} = 10 V dc. P_T = 275 ± 25 mW at T_A = room ambient as defined in the general requirements of MIL-STD-750. No heat sink or forced air cooling shall be permitted on the devices.

4.4.3 Group C inspection (all quality levels). Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Test condition E. Not applicable to surface mount devices.
C3	2006	Nonoperating, 30,000 G, X_1, Y_1, and Y_2 orientations.
C6	1027	I_F = 20 mA dc; V_{CE} = 10 V dc. P_T = 275 ± 25 mW at T_A = room ambient as defined in the general requirements of MIL-STD-750. No heat sink or forced air cooling shall be permitted on the devices. T = 1,000 hours.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input LED tests. These tests shall be performed with the output transistor terminals open.

4.5.3 Output transistor tests. These tests shall be performed with the input LED terminals open.

4.5.4 Isolation and coupling capacitance tests. The tests shall be performed between both input terminals 5 and 7 shorted together and the output terminals 1, 2, and 3 shorted together (see figure 1).

4.5.5 Monitored thermal shock (temperature cycling). One cycle of monitored temperature cycling shall be performed. All junctions shall be monitored for electrical continuity. Any discontinuity shall be cause for rejection of the device under test.

4.6 Internal visual (PRECAP) inspection (JANTXV level only). Internal visual inspection shall be performed in accordance with MIL-STD-750, methods 2072 and 2073 and as specified herein. Method 2072 shall be used for inspecting transistor die and completed assembly. Method 2073 shall be used for inspecting light emitting diode.

4.6.1 Gallium arsenide or gallium aluminum arsenide light emitting diode die inspection. The die shall be inspected under 100X (minimum).

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
(LED characteristics)						
Reverse current	4016	V_R = 2 V dc (see 4.5.2)	I_R		100	µA dc
Forward voltage	4011	I_F = 10 mA dc (see 4.5.2)	V_{F1}	0.8	1.3	V dc
(Transistor characteristics)						
Collector to emitter breakdown voltage	3011	I_C = 1.0 mA dc; I_B = 0, (see 4.5.3)	$V_{(BR)CEO}$	35		V dc
Collector to base breakdown voltage	3001	I_C = 100 :A dc; I_F = 0, I_E = 0 (see 4.5.3)	$V_{(BR)CBO}$	35		V dc
Emitter to base breakdown voltage	3026	I_E = 100 µA dc; I_C = 0, I_F = 0 (see 4.5.3)	$V_{(BR)EBO}$	4		V dc
Off-state collector current (phototransistor mode)	3041	Bias condition D; V_{CE} = 20 V dc; I_B = 0, I_F = 0 (see 4.5.3)	$I_{C(OFF)1}$		100	nA dc
Forward-current transfer ratio 2/	3076	V_{CE} = 5.0 V dc; I_C = 10 mA dc; I_F = 0 pulsed (see 4.5.3)	h_{FE}			
4N22, 4N22A 4N23, 4N23A 4N24, 4N24A				200 300 400		
(Coupler characteristics)						
On-state collector current 2/ (phtotransistor mode)	3041	V_{CE} = 5 V dc; I_F = 2 mA dc	$I_{C(ON)1}$			
4N22, 4N22A 4N23, 4N23A 4N24, 4N24A				0.15 0.2 0.4		mA dc mA dc mA dc

See footnotes at end of table.

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TABLE I. Group A inspection. Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
On-state collector current 2/ (phototransistor mode)	3041	V_{CE} = 5 V dc; I_F = 10 mA dc	$I_{C(ON)2}$			
4N22, 4N22A 4N23, 4N23A 4N24, 4N24A				2.5 6.0 10.0		mA dc mA dc mA dc
Collector to emitter saturated voltage	2071	I_F = 20 mA dc; Pulsed (see 4.5.1)	$V_{CE(sat)}$		0.3	V dc
4N22, 4N22A 4N23, 4N23A 4N24, 4N24A		I_C = 2.5 mA dc I_C = 5 mA dc I_C = 10 mA dc				
Subgroup 3						
High temperature operation:		T_A = +100°C				
(Transistor characteristics)						
Off-state collector current (phototransistor mode)	3041	Bias condition D; V_{CE} = 20 V dc; I_B = 0, I_F = 0 (see 4.5.3)	$I_{C(OFF)2}$		100	µA dc
(Coupler characteristics)						
On-state collector current 2/ (phototransistor mode)	3041	V_{CE} = 5 V dc; I_F = 10 mA dc	$I_{C(ON)3}$			
4N22, 4N22A 4N23, 4N23A 4N24, 4N24A				1.0 2.5 4.0		mA dc mA dc mA dc
(LED characteristics)						
Forward volltage	4011	I_F = 10 mA dc (see 4.5.2)	V_{F2}	1.0	1.2	V dc
Low temperature operation:		T_A = -55°C				

See footnotes at end of table

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TABLE I. Group A inspection. Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Input to output capacitance		$\lvert V_{IN} - V_{Out} \rvert$ = 0 (see 4.5.4) f = 1 MHz	C_{IO}		5	pF
Input to output internal resistance		$\lvert V_{IN} - V_{Out} \rvert$ = 1 kV dc (see 4.5.4)	R_{IO}	10^{11}		Ω
Rise time 2/ (phototransistor mode)		V_{CC} = 10 V dc; I_F = 10 mA dc; R_L = 100 ohms (see figure 3)	t_r			
4N22, 4N22A					15	µs
4N23, 4N23A					15	µs
4N24, 4N24A					20	µs
Fall time 2/ (phototransistor mode)		V_{CC} = 10 V dc; I_F = 10 mA dc; R_L = 100 ohms (see figure 3)	t_f			
4N22, 4N22A					15	µs
4N23, 4N23A					15	µs
4N24, 4N24A					20	µs
Subgroups 5 and 6 Not applicable						
Subgroup 7 Monitored thermal shock (temperature cycling)		See 4.5.5				

1/ For sampling plan, see MIL-PRF-19500.
2/ Unless otherwise specified, tests and conditions apply to all case outlines.

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TABLE II. Groups B and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Off-state collector current	3041	V_{CE} = 20 V dc, I_F = 0, condition D (see 4.5.3)	$I_{C(OFF)1}$		100	nA dc
2.	On-state collector current 4/ 4N22, 4N22A 4N23, 4N23A 4N24, 4N24A		V_{CE} = 5 V dc, I_F = 10 mA dc	$I_{C(ON)2}$	 2.5 6.0 10.0		 mA dc mA dc mA dc
3.	Collector to emitter saturation voltage 4N22, 4N22A 4N23, 4N23A 4N24, 4N24A		I_F = 20 mA dc I_C = 2.5 mA dc I_C = 5 mA dc I_C = 10 mA dc	$V_{CE(SAT)}$		0.3	V dc
4.	Off-state collector current	3041	V_{CE} = 20 V dc, I_F = 0, condition D (see 4.5.3)	$\Delta I_{C(OFF)1}$	±100 percent of initial value or 25 nA dc; whichever is greater.		
5.	On-state collector current 4/		V_{CE} = 5 V dc, I_F = 10 mA dc	$\Delta I_{C(ON)2}$ 5/	± 25 percent change from initial reading.		

1/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 2, and 3.
b. Subgroup 4, see table II herein, steps 1, 2, and 3.
c. Subgroup 5, see table II herein, steps 1, 2, 4, and 5.

2/ The electrical measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 2, and 3.
b. Subgroup 3, see table II herein, steps 1, 2, 4, and 5.
c. Subgroup 6, see table II herein, steps 4 and 5.

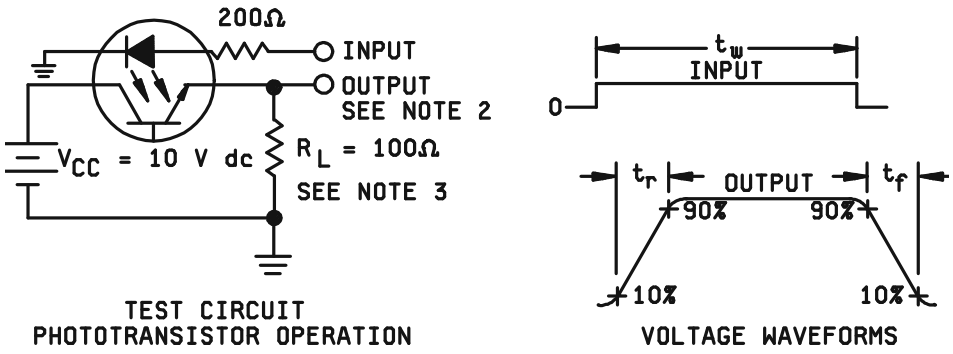
3/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 2, and 3 (JANS); and steps 1 and 2 (JANTX and JANTXV).
b. Subgroup 3, see table II herein, steps 2 and 3.
c. Subgroup 6, see table II herein, steps 4 and 5.

4/ Unless otherwise specified, tests and conditions apply to all case outlines.

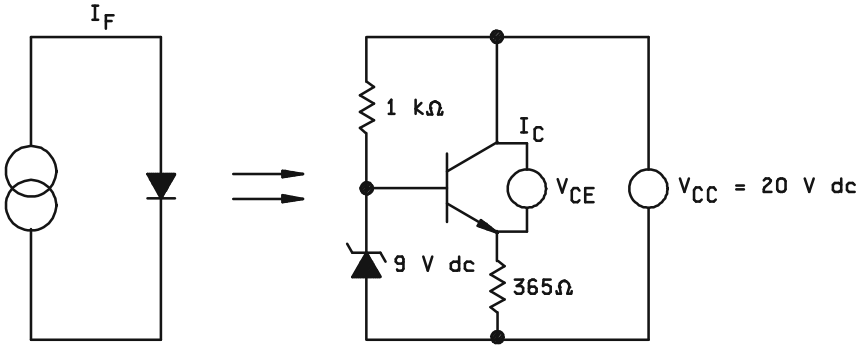
5/ Devices which exceed the group A limits for this test shall not be acceptable for shipment.

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NOTES:
1. The input waveform is supplied by a generator with the following characteristics: Z_{OUT} = 50 Ω, $t_r \leq$ 15 ns, duty cycle = 1 percent, t_W = 100 µs
2. The output waveform is monitored on an oscilloscope with the following characteristics: $t_r \leq$ 1 ns, R_{IN} = 1 MΩ, C_{IN} = 20 pF.
3. Adjust amplitude of input for: I_F = 10 mA dc.

FIGURE 3. Switching times.



NOTES:
1. I_F = 20 mA dc for operation life tests.
2. I_F = 40 mA dc for burn-in of TX and TXV.

FIGURE 4. Operation life test burn-in circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2) When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Air Force - 17
NASA - NA

Review activities:
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

(Project 5980- 0016)

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/486E	**2. DOCUMENT DATE (YYMMDD)** 970610

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, COUPLER, OPTOELECTRIC. SOLID STATE; TYPES 4N22, 4N22A, 4N23, 4N23A, 4N24, 4N24A, 4N22U, 4N22AU, 4N23U, 4N23AU, 4N24U, 4N24AU; JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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